

The Cooperative Elementary School: Effects on Students' Achievement, Attitudes, and Social Relations

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This article reports the results of a 2-year study of the cooperative elementary school model which used cooperation as an overarching philosophy to change school and classroom organization and instructional processes. The components of the model include: using cooperative learning across a variety of content areas, full-scale mainstreaming of academically handicapped students, teachers using peer coaching, teachers planning cooperatively, and parent involvement in the school. After the first year of implementation, students in cooperative elementary schools had significantly higher achievement in reading vocabulary. After the second year, students had significantly higher achievement in reading vocabulary, reading comprehension, language expression, and math computation than did their peers in traditional schools. After 2 years, academically handicapped students in cooperative elementary schools had significantly higher achievement in reading vocabulary, reading comprehension, language expression, math computation, and math application in comparison with similar students in comparison schools. There also were better social relations in cooperative elementary schools, and handicapped students were more accepted socially by their nonhandicapped peers than were similar students in traditional schools with pull-out remedial programs. The results also suggest that gifted students in heterogeneous cooperative learning classes had significantly higher achievement than their peers in enrichment programs without cooperative learning.

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Cooperative learning methods have been researched for more than 20 years and have been rapidly increasing in use over the past decade. However, until recently almost all applications of cooperative learning methods have been implemented teacher by teacher, usually in one subject at a time, and few studies have involved durations of more than a semester (see Slavin, 1994). Cooperative learning methods can certainly be justified as additions to teachers' instructional repertoires in traditionally organized schools, but the potential for cooperative approaches to serve as the organizing principle for the entire school and as the core instructional element has not been assessed. Could cooperative learning be used on a broad scale in many subjects and over extended time periods to fundamentally change the organization of schools and classrooms? Would cooperative learning methods still be effective if they became the primary mode of instruction in schools, and would they maintain their effectiveness over time? Would schoolwide use of cooperative principles enhance the school's potential to successfully mainstream learning disabled students? These were the principal questions addressed in the study reported here.

The goal of this research was to use cooperative learning as a way to change classrooms and learning activities. Cooperative learning was seen as a vehicle for increasing students' active involvement in learning activities and having students take more responsibility for managing their own instruction, thus allowing the teacher more opportunities to have academic interactions with individuals and small groups.

At the same time, we attempted to use cooperation as a philosophy to change the school and classroom organization and the way members of the school community interacted. Peer coaching was used as a form of cooperation where teachers would help one another improve their teaching (Joyce, Hersh, & McKibbin, 1983). Similarly, teachers and the principal were encouraged to collaborate in planning instruction and to develop interdependence within the faculty so they would take collective responsibility for the learning of all students.

Research on Cooperative Learning

In a review of the research on cooperative learning, Slavin (1994) focused on studies of at least 4 weeks' duration that compared cooperative learning and control classes on objectives pursued equally by both over periods of at least 4 weeks. The review concluded that for cooperative learning to have an effect on achievement two conditions were essential: group goals and individual accountability. Group goals motivate students to help their group mates learn. They develop positive interdependence between the individuals in the group, giving them a reason to cooperate in a meaningful fashion (Deutsch, 1949; Johnson & Johnson, 1989; Stevens, 1994). Individual accountability, a measure of each students' learning, increases the probability that all students will learn and reduces the potential for a *free rider effect*, where a student does little and relies on others in the group to accomplish the goal (Johnson & Johnson, 1989; Slavin, 1994; Stevens, 1994).

However, these two elements must be linked to make cooperative learning processes effective in increasing learning. The attainment of the group goal must depend on the individual learning of all group members. In this way, the group's success depends on successful learning by everyone and produces positive interdependence. When this kind of interdependence is developed, students perceive their role to be to help their groupmates *learn*, not to simply help them complete a single group task. The research of Webb (1985) and others has shown that achievement gains in cooperative learning depend on the giving and receiving of elaborate explanations of concepts and skills and that giving and receiving answers without explanations are negatively related to achievement gain. Explanations force students to evaluate, integrate, and elaborate on knowledge and thus improve their understanding or depth of processing of what is being learned (Brown & Campione, 1986; Wittrock, 1986). It is hypothesized that the use of group goals based on individual accountability motivates students to provide high-quality explanations to one another and to avoid simply giving answers. Reviewers of the extensive literature on cooperative learning and student achievement have generally agreed that group goals based on individual accountability measures are essential for achievement effects (Davidson, 1985; Johnson & Johnson, 1989; Newmann & Thompson, 1987; Slavin, 1994). Cooperative learning programs that incorporate these elements consistently increase student achievement more than control methods. Slavin reports that a total of 52 such studies had a median effect size¹ on achievement measures of $+ .32$. In contrast, the median effect size for methods lacking group goals and individual accountability was only $+ .07$ across 25 studies.

Cooperative learning has been found to have a positive impact on many variables other than achievement, including intergroup relations, self-esteem, attitudes toward class and school, and ability to work collaboratively with others (Johnson & Johnson, 1989; Slavin, 1994). These noncognitive outcomes, however, do not seem to depend on the use of group goals based on individual accountability measures.

There has been some question about whether the effects of cooperative learning can be maintained in longer implementations and when cooperative learning is used throughout the school day rather than in one subject at a time. Critics (e.g., Bossert, 1989) have sometimes questioned whether the effects of cooperative learning may be due to novelty or Hawthorne effects and whether intensive, long-term use of cooperative learning programs might be less effective. On the other hand, it may be that, as students experience cooperative methods in many subjects and for longer periods, their increasing skills in working cooperatively and a perception that cooperation leads to success could make widespread and long-term use of cooperative learning more effective than more time-limited implementations.

Year-long studies of cooperative learning programs using group goals based on individual accountability have generally been as successful as shorter studies in enhancing student achievement (e.g., Slavin & Karweit, 1984; Slavin, Leavey, & Madden, 1984; Stevens, Madden, Slavin, & Farnish,

1987). The few multiyear studies that have been done have found effects that vary according to the methods used. In a 2-year study of Cooperative Integrated Reading and Composition (CIRC), Stevens and Slavin (1995) found that a cooperative learning reading and language arts program that used group goals based on individual accountability had significant positive effects on students' achievement in reading and language arts. Similarly, a 3-year study of CIRC in bilingual classes by Hertz-Lazarowitz, Ivory, and Calderon (1993) found positive effects on reading and language performance. Several multiyear studies of Success for All, an elementary school restructuring model that makes extensive use of cooperative learning methods, have also found substantial positive effects on students' reading performance (Slavin et al., in press).

In contrast, a multiple-year study by Talmage, Pascarella, and Ford (1984) studied a 3-year staff development emphasizing cooperative learning methods developed by Sharan and Sharan (1976) in which students were instructed to break group tasks into subtasks and were graded based on an evaluation of the group's product. This program produced small positive effects on student achievement in reading and language arts. A study by Solomon, Watson, Schaps, Battistich, and Solomon (1990) implemented cooperative learning methods over a 5-year period, starting with children in kindergarten and continuing with the same cohort through the fourth grade. The study found no effects on achievement as compared to control schools, but there were positive effects on prosocial behavior, social problem solving, and adherence to democratic values. It should be noted, however, that the latter two studies used forms of cooperative learning that emphasized the production of a group product. While the models in these studies used individual accountability (e.g., grades), *group* success was not dependent on individual learning performance, and there were no formal group goals. The group products did not require each student to participate equally, and there was no way to hold each student accountable for learning during the process.

Given the quantity of research that documents the potential for cooperative learning to have a positive impact on students' achievement in a variety of subject areas, as well as the potential social and attitudinal benefits, it seems that cooperative learning may be a viable theme for changing the organization of an entire school. Therefore this study attempted to reorganize the school and classrooms in the school by using cooperative learning across the curriculum and using it as a theme to better integrate instruction between special and regular education.

The Cooperative Elementary School

The cooperative elementary school model uses cooperation as a philosophical and practical approach to changing classroom and school organization, classroom processes, and learning activities to provide students with more active learning experiences, equal access to learning by all students, and a

more supportive social environment for the students and teachers (Slavin, 1987). The elements were the following:

1. Widespread use of cooperative learning in academic classes.
2. Mainstreaming learning disabled students in regular education.
3. Teachers coaching one another.
4. Teachers collaborating in instructional planning.
5. Principal and teachers collaborating on school planning and decision making.
6. Principal and teachers encouraging active involvement of parents.

We describe each of the elements and the theoretical and research rationale for them below.

Widespread Use of Cooperative Learning in Academic Classes

A major emphasis of the cooperative elementary school design is the use of cooperative learning methods in many subjects and grade levels. Cooperative learning changes the classroom organization and processes to make learning more active for students and increase students' responsibility for their learning activities. Similarly, a cooperative learning classroom organization allows teachers more opportunities to meet instructionally with small groups and individual students because it frees teachers from many of their management responsibilities, which are assumed by the students and their learning groups (Johnson & Johnson, 1981; Slavin, 1994).

Teachers in cooperative elementary schools are given training, materials, follow-up, and assistance to use the Johns Hopkins cooperative learning models (Slavin, 1986). For the present study, particular emphasis was placed on the two comprehensive programs designed to accommodate student diversity: Cooperative Integrated Reading and Composition, or CIRC, (Hertz-Lazarowitz et al., 1993; Stevens et al., 1987; Stevens & Slavin, 1995) and Team Assisted Individualization-Mathematics, or TAI (Slavin et al., 1984, 1986). These programs included instructional materials, teachers' manuals, and other supports to enable teachers of Grades 2–6 to use them as their primary approaches to the teaching of reading, language arts/writing, and mathematics, respectively. Research on TAI (e.g., Slavin et al., 1984, 1986; Slavin & Karweit, 1985) and CIRC (Stevens et al., 1987; Stevens & Slavin, 1995) has found these methods to be among the most effective of all cooperative learning programs for accelerating student achievement.

Mainstreaming Learning Disabled Students in Regular Education

Another key element of the cooperative school model is the intention to mainstream learning disabled students into regular education. There has been a great deal of discussion and research on mainstreaming special education students since the passage of PL 94-142, the Education of All Handicapped Act in 1975. Yet the question of the efficacy of special education and remedial pull-out programs for mildly handicapped students is certainly not a new

one (Lloyd & Gambatese, 1991). Separate special education programs have been depicted as expensive, ineffective, and poorly integrated with regular education instruction (cf. Johnston, Allington, & Afflerbach, 1985; Stainback & Stainback, 1984).

Current research on mainstreaming has been strongly influenced by advocates of educating students with handicaps in the least restrictive environment possible (Reynolds, 1989; Will, 1986). However, the research on mainstreaming to date is less than conclusive and has often raised more questions than answers (Fuchs & Fuchs, 1991; Lloyd, Crowley, Kohler, & Strain, 1988). Research has suggested that simply placing students with learning disabilities in regular classes is not sufficient to produce significantly better achievement (Marston, 1987). Similarly, regular classroom teachers often feel unprepared for or incapable of providing adequate instruction for learning disabled students who are added to their classes (Gersten, Walker, & Darch, 1988). It seems that effective mainstreaming may require restructuring schools to provide additional support for the instructional needs of learning disabled students in the regular classroom setting (Stainback & Stainback, 1991).

How can schools effectively mainstream academically handicapped students? One alternative may be cooperative learning. Previous research has shown the potential benefits of cooperative learning processes for helping teachers mainstream special education students and for improving their social relations and academic achievement (Johnson, Johnson, Waring, & Maruyama, 1986; Madden & Slavin, 1983; Slavin, 1984; Slavin, Stevens, & Madden, 1988; Slavin & Stevens, 1991; Stevens & Slavin, 1991, 1995). However, there is a need for more research; not all of the programs that mainstreamed learning disabled students through cooperative learning have shown significant effects on students' achievement (Fuchs & Fuchs, 1991; Jenkins et al., 1994; Lloyd et al., 1988; Stevens & Slavin, 1991). Furthermore, questions remain about the long-term academic and social effects of mainstreaming through cooperative learning and about the impact of using cooperative learning throughout the day in a variety of subjects as a means of mainstreaming learning disabled students.

The previous research that effectively used cooperative learning to mainstream learning disabled students typically discontinued remedial pull-out programs. Instead, the special education teacher team taught in the regular classroom along with the regular education teacher (e.g., Stevens et al., 1987; Stevens & Slavin, 1995). In this way, it was possible to give the learning disabled students the additional instructional support they required.

In the cooperative elementary school model, learning disabled students receive all of their instruction in the regular classroom. During reading and/or math periods, the special education teacher team teaches with the regular classroom teacher using CIRC or TAI. In both of these methods, students work in heterogeneous learning teams, but receive instruction in relatively homogeneous teaching groups drawn from the various teams. The teaching groups are often composed of both learning disabled and nondisabled stu-

dents. Typically, the regular education teacher provides the initial instruction for the students. The special education teacher provides follow-up on and extensions of the regular instruction. For example, the special education teacher often provides more support for the learning disabled students by giving them additional instruction related to what was covered in the teaching group or by guiding them and giving them feedback during their initial practice activities.

Mainstreamed learning disabled students are integrated into the cooperative learning teams within the class, distributing the learning disabled students among the teams with the nonhandicapped students. The mainstreamed students interact with other students on academic tasks as they work in cooperative learning teams, giving the learning disabled students all of the potential cognitive advantages of cooperative learning described previously. The cooperative learning processes also provide a way for special education students to become socially mainstreamed through academic interactions with their nonhandicapped classmates.

Organizational Support

Other elements of the cooperative elementary school model are used to create more support within the schools for the changes being made in the school and classroom organization aspects of the model. In order to fully implement widespread use of cooperative learning and full-scale mainstreaming of learning disabled students, certain school environment elements are also considered. Those elements include: peer coaching, teacher collaboration in instructional planning, principal-teacher collaboration in administrative planning, and parent involvement.

Peer coaching. One major difficulty in using an innovative program that changes how classrooms are organized and how teachers teach is the quality of the implementation (Showers, 1987). While some implementation issues may be addressed through the specificity of the training and periodic follow-up, implementation continues to be problematic. Peer coaching, where teachers observe and give feedback to one another, is an approach that has successfully produced higher implementation of innovative programs (Joyce & Showers, 1980, 1981; Showers, 1987; Williamson & Russell, 1990) and specific teacher behaviors (Sparks, 1986).

Peer coaching provides more opportunities for classroom teachers to get specific and nonthreatening feedback as they use new programs in their classrooms. Through coaching one another, teachers develop expertise within the school, rather than simply being the recipients of external expertise. This helps them develop more ownership of the innovation and more confidence as they try the new strategies involved (Sparks & Burder, 1987). At the same time, peer coaching provides a vehicle for professional dialogue about teaching that also seems to increase feelings of collegiality within the school (Sparks & Burder, 1987).

In the cooperative elementary school model, teachers are given many opportunities to visit one another's classes and provide support and feedback

to one another. Teachers who have successfully implemented a given cooperative learning method serve as peer coaches for those who will receive training later. The principals encourage peer coaching by offering to teach classes for the coaches when they are observing their colleagues.

Teacher collaboration in instructional planning. As part of the cooperative elementary school's philosophical approach to using cooperation as the central element in school change, we wanted teachers to collaborate with one another on substantive issues related to instructional planning and implementation. Little (1982) found that continuous professional development can be found in schools that promote substantive dialogue among teachers about teaching and curriculum planning and development. Similarly, Johnson and Johnson (1987) suggested that teacher collaboration can help improve teachers' competence in implementing innovations.

In the cooperative elementary schools, we promoted collaborative teacher dialogue within the context of implementing the elements of the model. Teachers were given time to plan common strategies, instructional content, and activities in the forum of grade-level meetings. This provided the teachers an opportunity to coordinate academic content across classes and facilitated the sharing of instructional materials. Teachers also used their planning time to develop classroom strategies for covering content effectively and resolving management problems.

Principal and teacher collaboration in administrative planning. Another key element of the cooperative elementary school model was to develop collaboration between the administrators and teachers to further support the process of change. The building steering committee was made up of the principal, representatives from each of the grade levels, special services, and other faculty members. The goal of the committee was to develop goals for the school and to discuss progress toward the goals periodically. The committee provided a forum for discussion of broad issues related to school management and policy as well as integration of instructional goals and practices across grade levels. While the principal remained the instructional leader in the school, and was ultimately responsible for making schoolwide decisions, the steering committee provided valuable input in the decision-making process.

Parent involvement. The cooperative elementary school model encouraged parent participation in school activities to develop a sense of joint responsibility for children's success in school. Parents and the community were introduced to the project as it was initiated. They were given a theoretical rationale for the project and an explanation of each of the components of the cooperative elementary school. The parents were encouraged to monitor students' educational progress at home. The parents were also presented with the school's expectations concerning homework and independent reading, and the parent's role in making sure that those were accomplished. The cooperative elementary schools also maintained regular contact with parents through regularly scheduled teacher-parent conferences, PTA meetings, and a school newsletter. The latter kept parents informed of events at the school,

changes in school policy or programs, and notable successes of students, classes, or teachers.

Method

Sample and Design

The sample consisted of 1,012 students in second through sixth grades in five elementary schools of a suburban Maryland school district. Twenty-one classes in the two treatment schools were matched with 24 classes in the three comparison schools on mean California Achievement Test scores for Total Reading, Total Language, and Total Mathematics. There was also an attempt to control for ethnic and socioeconomic background of the students by selecting comparison schools from the same or similar neighborhoods (see below). The student populations of each school ranged from 4% to 15% minority students ($M = 7.3\%$), and from 2% to 20% ($M = 10.2\%$) disadvantaged students (i.e., those receiving free or reduced-price lunch). The schools were all located in predominantly working-class neighborhoods. Approximately 9.3% of the five schools' student populations were identified as learning disabled, ranging from 7% to 12% in each school.

Selection of Schools

Treatment schools. Because the cooperative elementary school required schoolwide change and encouraged teacher participation in decision making, we felt it was important to have teachers involved in determining whether their school would participate in this project. Prior to beginning the project, we approached four schools in the Maryland school district that agreed to participate in the project. At each school, we presented a brief overview of the project, the duration, and the potential benefits. After the authors answered the teachers' questions and concerns, the staff voted on participating in the proposed project. Only if 75% of the faculty agreed to participate in all elements of the project was the school selected; of the four schools to whom presentations were made, two voted overwhelmingly to participate.

Comparison schools. The comparison schools were selected from school district records of average student achievement, ethnicity, and socioeconomic background. These data were used to select comparison schools that were well matched with the two treatment schools on these independent and demographic variables. Schools that were approached for the study but did not volunteer to participate were not included as comparison schools.

Treatments

Treatment schools. The treatment schools adopted the cooperative elementary school model, as described previously. Each component was gradually phased in during the first year of the study. Prior to implementation of a component, the faculty was given training or an introduction to the component that included the rationale and goals as well as detailed teachers' manuals

for each instructional program. During the school year, members of the research staff observed classes, participated in meetings with teachers, and observed steering committee meetings to facilitate implementation of the program components.

Comparison schools. The comparison schools continued using their regular teaching methods and curriculum. In reading, these methods consisted of the use of two or three ability-based reading groups. A basal series was used with workbooks, worksheets, and teacher-prepared materials as follow-up activities. The teachers also integrated literature into reading instruction by reading two or three novels a year. In language arts, the teachers used whole-class instruction and usually used a published language arts series as well as teacher-made activities in writing and language usage. In mathematics, the teachers used a district-adopted mathematics text. Typically, mathematics involved whole-class instruction with follow-up activities from the text or generated by the teacher. While the comparison teachers occasionally used various kinds of group work, none of them used structured cooperative learning that involved group goals based on group members' individual learning performances.

The comparison schools did not implement most of the other components of the cooperative elementary school model. Specifically, the comparison schools did not use an in-class model for mainstreaming special education students. They did not use peer coaching or regularly use cooperative planning between teachers. The comparison schools did have school improvement teams made up of administrators and faculty, much like the cooperative elementary school's building-level steering committee, but typically the school improvement team met much less frequently (e.g., 2 times a semester rather than 2 times a month).

Allocated time and curricula. Both the treatment and comparison schools allocated the same amount of time to reading, language arts, and mathematics instruction daily, in compliance with school district guidelines. In reading, all of the schools used similar basal reading programs selected from an adoption list provided by the school district. However, the treatment schools did not use adjunct materials (i.e., workbooks) provided by the basal publisher, whereas the comparison schools typically did use them. Similarly, in language arts both the treatment and comparison schools used a district-adopted language arts program, but in the cooperative elementary schools this program was used as supplemental material during CIRC writing, rather than as the central curriculum. In mathematics, the comparison schools used the district-adopted textbook, whereas the treatment schools used the TAI curriculum, using the district text for application and problem solving activities only.

Implementation of the Model Components

Steering committee. The first component of the cooperative elementary school to be implemented was the building-level steering committee. It was important to establish this committee because it would act as the main vehicle

for communication between teachers, school administration, and research staff. The steering committee was organized prior to the start of the school so that it could help resolve problems that might occur as we began implementing the other components of the model.

Cooperative learning programs. We gradually phased in the implementation of the cooperative learning programs during the first year of the project. The goal was to have cooperative learning used by all of the teachers across a variety of curriculum areas, with specific emphasis on the CIRC program in reading and language arts and the TAI program in mathematics. The teachers also were trained in Jigsaw II, Teams-Games-Tournaments (TGT), and Student Teams Achievement Division (STAD), which are generic cooperative learning models that can be used in a variety of content areas, such as social studies, spelling, or science (see Slavin, 1986). The training for each program included explanation of the processes and the rationale behind them, and a detailed manual on how to use the program in a classroom. Trainers also provided teachers with simulated demonstration lessons for each program.

The initial training occurred in August, with training on subsequent cooperative learning programs taking place at roughly 2-month intervals so the teachers were not overwhelmed by too many new programs at the same time. Initially, the teachers were trained in the CIRC program's reading component because we felt it would be easier to start the year with a new reading program than it would be to reorganize reading instruction during the year. Also, CIRC was considered the most complex program, so teachers would have the most time to master it.

Following training, the research staff observed the teachers as they used the programs in their classrooms. We gave the teachers feedback on their implementation to improve the quality of their instruction and maintain fidelity to the program. Teachers also coached one another as they started using the cooperative learning programs.

Subsequent training was planned in collaboration with the building steering committee. They provided valuable feedback on whether the faculty was ready to take on another new program and which program they preferred to implement next. By March of the first year of the study, all of the curricular elements of the program were in place. All of the teachers were using CIRC and TAI as their reading, language arts, and mathematics curricula. In a typical week, the classes used the instructional procedures prescribed in those models for least 4 days. (Teachers occasionally used one day a week for other kinds of activities like performing a play, watching a movie; engaging in reading, writing, or mathematics activities related to other subjects like social studies and science, or engaging in enrichment activities like field trips.)

All of the teachers were trained in the more generic cooperative learning models (Jigsaw II, TGT, and STAD). These model are designed and used for specific activities in a variety of subjects like spelling, social studies, and science. Unlike the curriculum-specific models of cooperative learning, the generic models are designed to be an adjunct to instruction, and, to a great

extent, they are different ways to accomplish the same goal. As such, the teachers used them much more irregularly and may not have used each of them. However, all of the teachers were observed a number of times using at least one of the generic models during instructional activities in subjects other than reading, language arts, and mathematics.

One aspect of the cooperative elementary school, mainstreaming learning disabled students, was only partially implemented the first year. Due to scheduling difficulties, the schools were able to mainstream only about 60% of their learning disabled students.

Elements of the cooperative elementary school. During the second year of the study, all the elements of the cooperative elementary school were in use. The teachers were using cooperative learning in a variety of subject areas, and the students were involved in cooperative learning on a daily basis. Special education pull-out remediation for learning disabled students had been discontinued, and special education teachers provided remedial instruction to them in the regular classroom in team-teaching arrangements. The learning disabled students were also integrated with nonlearning disabled students on heterogeneous learning teams. The teachers within and sometimes across grades collaborated in planning instruction, and teachers at a grade level met at least weekly for instructional planning. The building steering committee met at least monthly to plan and evaluate the school's progress in meeting their goals. Finally, a biweekly newsletter was published by the school to keep the parents and community informed about the activities at the school, and parents were provided with periodic parent-faculty meetings about the nature of and events in the cooperative elementary school.

Measures

Achievement pretests. We used standardized test scores from district records to match classes on entering ability and as a covariate in the statistical analyses for the entire population. The scores used were the Total Reading, Total Language, and Total Mathematics scores from the California Achievement Test, Form C, which were given in the fall of the first year of the study. The pretest scores were transformed to z scores separately for each grade, so that the data could be analyzed by grade level and collapsed across grade levels if there were no grade-by-treatment interactions.

Achievement posttests. In the spring of the first and second years of the study, teachers administered the reading, language, and mathematics subtests of the California Achievement Test, Form E. The subtests of California Achievement Test have reliabilities (internal consistency) ranging from .80 to .95, with a median .91 reliability (CTB/McGraw-Hill, 1986). Raw scores from the subtests were transformed to z scores separately for each grade to enable us to combine scores across grades if there were no grade-by-treatment interactions.

Attitude measures. The students were given an attitude measure as a pre- and posttest. Pretests were given in the fall of the first year of the project, and posttests were given in the spring of the second year. The attitude

measure asked students to rate their attitude toward and their perceived ability in reading, language arts, and mathematics. For each subject, the students were also asked to rate their interest and their ability on a 3-point scale ("I like it a lot"/"I don't like it" and "I am really good at this"/"I am not very good at this"). The attitude measure had alpha reliabilities of .35 for the premeasure and .33 for the postmeasure. For the perceived ability measure, the alpha reliabilities were .32 and .37, respectively.

Social relations measure. The students were given a measure of their social relations as a pre- and postmeasure at approximately the same time they were given the other pre- and posttests. The social relations measure asked students to list the names of their friends in the class (Moreno, 1953). The results were analyzed by comparing the average number of friends listed by treatment and control students. The test-retest reliability for the number of friends was .57 for this population. For learning disabled students, the social relations measures were reanalyzed to determine the number of times each learning disabled student was selected as a friend by his/her nonhandicapped peers. This second analysis provided a measure of social acceptance of mainstreamed learning disabled students.

Results

Analyses

The analyses were somewhat complicated by the use of intact classes and schools. The students were nested within classes, classes were nested within schools, and schools were nested within treatment. Because of the nested nature of the design, each level of the hierarchy potentially influences the processes at the lower levels (Raudenbush & Bryk, 1988). In an attempt to disentangle the effects of schools or teachers from the effects of the treatments on the student outcomes, we used the hierarchical linear model (HLM; Bryk, Raudenbush, Seltzer, & Congdon, 1988). Raudenbush and Bryk (1988) provide a thorough discussion of the use of HLM with multilevel data.

Since the study lasted 2 years, the posttest data were analyzed and reported separately for each year. The initial sample of the five schools included 1,012 students. Of those, 873 students had pretest data available in the district's records and were in the study for the entire 2 years. Approximately 13.7% of the original students moved or otherwise did not stay in the participating school for 2 years. (In the treatment schools, 11% of the students remained in the school for the entire 2 years.)

Standardized test scores that were converted into z scores were used in the achievement analyses, with the appropriate pretest (Total Reading, Total Language, or Total Mathematics) used as a covariate to increase the power of the analyses. The test of the homogeneity of regressions indicated no significant violations of that assumption, all $t < 1.0$. Initially, grade-by-treatment analyses were conducted to determine if the treatment had differential effects on achievement, attitudes, or social relations at different grade levels. No grade-by-treatment analyses were significant, so the data

were collapsed across grade levels to simplify the analyses and discussion of the results.

Achievement pretests. As noted previously, the treatment and comparison classes were matched on initial achievement on the California Achievement Test. There were no significant pretest differences on Total Reading, $t < 1.0$, or Total Language, $t < 1.0$ (see Table 1). There were, however, significant pretest differences on Total Mathematics, $t = 2.02, p = .05$, favoring the comparison students.

Posttests after the first year. The results from the first-year analyses indicated significant differences in favor of the treatment group on reading vocabulary, $t = 2.14, p < .05$ (see Table 1). The magnitude of the effect, however, was relatively small; the effect size was $+ .17$ standard deviations. There were no significant differences on the measures of reading comprehension, $t = 1.63$; language mechanics, $t < 1.0$; language expression, $t < 1.0$;

Table 1
Students' Achievement: Means, Standard Deviations, Analyses, and Effect Sizes (Collapsed Across Grades)

Measure	CES			Comparison			<i>t</i>	Effect size ¹
	<i>M</i>	(<i>SD</i>)	HLM-fitted <i>M</i>	<i>M</i>	(<i>SD</i>)	HLM-fitted <i>M</i>		
Pretests								
Reading	-.03	(.99)		.04	(1.00)		<1.0	-.07
Language	-.01	(.98)		.00	(1.00)		<1.0	-.01
Mathematics	-.09	(.97)		.08	(1.02)		2.02*	-.17
Year 1								
Posttests								
Read. voc.	.04	(.99)	.08	-.14	(1.01)	-.09	2.14*	+ .17
Read. comp.	.03	(1.01)	.08	-.03	(.99)	-.05	1.63	+ .13
Lang. mech.	-.01	(.99)	.00	-.02	(1.00)	.01	<1.0	-.01
Lang. expr.	.04	(1.01)	.04	-.04	(.99)	-.04	<1.0	+ .08
Math comp.	.02	(.99)	.06	-.01	(1.00)	-.06	1.34	+ .12
Math appl.	-.07	(.98)	-.02	.08	(1.01)	.03	<1.0	-.05
Year 2								
Posttests								
Read. voc.	.05	(.98)	.10	-.04	(1.01)	-.11	3.04**	+ .21
Read. comp.	.08	(.98)	.15	-.07	(1.01)	-.13	3.62**	+ .28
Lang. mech.	.03	(.97)	.05	-.02	(1.02)	-.05	1.16	+ .10
Lang. expr.	.10	(.96)	.11	-.09	(1.03)	-.10	2.93**	+ .21
Math comp.	.03	(1.00)	.15	-.04	(.99)	-.14	.377**	+ .29
Math appl.	-.03	(1.01)	.04	.01	(1.01)	-.06	1.24	+ .10
<i>N</i> (students)	411			462				
Number of classes	21			24				

¹Effect size equals the difference of the HLM-fitted means divided by the control group standard deviation.

* $p < .05$.

** $p < .01$.

math computation, $t = 1.34$; and math application, $t < 1.0$. The means, HLM-fitted means, standard deviations, and effect sizes are presented in Table 1. The HLM-fitted means are equivalent to adjusted means and are computed from the base and gamma produced by the HLM analyses and the orthogonally coded comparison coefficient. The HLM mean = base + (G $\times$ C), where G is the gamma and C is the comparison coefficient.

It should be noted that the HLM analysis is analogous to a conservative class-level analysis. Individual-level analyses showed significantly positive effects of the model on reading vocabulary, reading comprehension, and math computations.

Posttests after the second year. The second year HLM analyses yielded significant effects favoring the treatment group on reading vocabulary, $t = 3.04$, $p < .01$; reading comprehension, $t = 3.62$, $p < .01$; language expression, $t = 2.93$, $p < .01$; and math computation, $t = 3.77$, $p < .01$ (see Table 1). There were no significant effects on language mechanics, $t = 1.16$, or math applications, $t = 1.24$. The magnitude of the significant effects ranged from .21 to .29 standard deviations.

Attitude measures. There were no significant pre- or postmeasure differences on students' attitudes toward reading, language arts, or mathematics. Similarly, there were no premeasure differences on students' perceived ability in reading, language arts, or mathematics. After 2 years, students in the cooperative elementary school did have higher perceived ability in reading ($t = 2.02$, $p < .05$) and language arts ($t = 2.99$, $p < .01$). The magnitude of these significant effects was +.20 and +.26, respectively (see Table 2).

Social relations. The social relations measure examined the number of friends listed by students in the treatment conditions at the beginning of the study and the number listed 2 years later at the end of the study. There were no significant differences on the premeasure, $t < 1.0$. On the postmeasure, the students in the treatment schools listed significantly more friends than did students in the comparison schools, $t = 3.92$, $p < .01$ (see Table 3). The magnitude of the significant effect was +.42 standard deviations.

Effects for learning disabled students after the first year. The pretests indicated that the learning disabled students in the cooperative elementary schools and in the comparison schools had similar levels of initial achievement. There were no significant pretest differences on standardized measures of their academic ability in reading, $F = 1.33$; language arts, $F = 3.19$; or mathematics, $F < 1.0$. After the first year, the achievement results for learning disabled students were somewhat similar to those of the whole population. The analysis of variance results indicated no significant differences between the achievement of learning disabled students in the cooperative elementary school and those in the comparison schools; however, learning disabled students in the cooperative elementary school did have slightly better achievement scores on all measures (see Table 4).

Effects for learning disabled students after the second year. After the second year, the differences in achievement for learning disabled students again followed the pattern of the differences found in the entire population.

Table 2
Students' Attitude, Means, Standard Deviations, and Effects

	CES		Comparison		<i>t</i>	Effect size ¹
Attitude toward reading						
Pretest	1.47	(.61)	1.55	(.63)	1.79	−.12
Posttest	1.70	(.69)	1.71	(.68)	<1.0	−.01
Attitude toward language arts						
Pretest	1.80	(.69)	1.74	(.74)	1.44	+ .08
Posttest	1.82	(.76)	1.78	(.74)	<1.0	+ .05
Attitude toward mathematics						
Pretest	1.50	(.69)	1.43	(.61)	1.52	+ .11
Posttest	1.48	(.65)	1.42	(.69)	1.37	+ .09
Perceived reading ability						
Pretest	1.58	(.56)	1.62	(.60)	<1.0	−.07
Posttest	1.72	(.59)	1.61	(.55)	2.02*	+ .20
Perceived language arts ability						
Pretest	1.73	(.65)	1.68	(.65)	1.31	+ .08
Posttest	1.74	(.65)	1.58	(.62)	2.99**	+ .26
Perceived math ability						
Pretest	1.39	(.66)	1.47	(.62)	1.22	−.12
Posttest	1.45	(.63)	1.51	(.67)	1.11	−.07
<i>N</i> (students)	411		462			
Number of classes	21		24			

¹Effect size equals the difference in treatment means divided by the control standard deviation.

* $p < .05$.

** $p < .01$.

There were significant posttest differences in favor of the learning disabled students in the cooperative elementary school on reading vocabulary, $F = 13.48$, $p < .01$; reading comprehension, $F = 14.39$, $p < .01$; language expression, $F = 11.41$, $p < .01$; math computation, $F = 10.77$, $p < .01$; and math application, $F = 3.75$, $p < .05$ (see Table 4). There were no significant differences in language mechanics, $F < 1.0$. The effect sizes for the significant effects were much larger than those found for nonhandicapped students, ranging from +.35 to +.85.

Initially, there were no significant differences between the treatment groups on learning disabled students' social relations, $F < 1.0$, as measured by the number of friends they selected. On the postmeasure, learning disabled students in the treatment schools listed significantly more friends than did their counterparts in the comparison schools, $t = 3.42$, $p < .01$ (see Table 3). This was a substantial difference, with an effect size of +.86. The social relations measure was also used to assess the level of the social acceptance of mainstreamed learning disabled students, as measured by the number of times learning disabled students were selected as friends by their nonhandicapped classmates. Initially, there were no significant differences, $t < 1.0$. On the postmeasure, learning disabled students in the treatment schools

Table 3
**Classroom Social Relations, Means, Standard Deviations,
 and Effect Sizes**

	CES	Comparison	<i>t</i>	Effect size ¹
Number of friends				
Premeasure	5.91 (3.67)	5.89 (3.59)	<1.0	+ .01
Postmeasure	8.71 (3.82)	7.12 (3.62)	3.92**	+ .42
<i>N</i> (students)	411	462		
Gifted students				
Number of friends				
Premeasure	5.74 (2.23)	5.66 (2.54)	<1.0	+ .03
Postmeasure	8.92 (3.01)	7.59 (2.89)	2.64**	+ .46
<i>N</i> (students)	46	61		
Special education students				
Number of friends				
Premeasure	5.03 (2.81)	5.10 (3.04)	<1.0	- .02
Postmeasure	9.11 (4.27)	6.29 (3.29)	3.42**	+ .86
Number of times picked by nonspecial education students				
Premeasure	2.83 (1.76)	2.78 (1.54)	<1.0	+ .03
Postmeasure	5.83 (2.39)	3.81 (2.24)	4.33**	+ .90
<i>N</i> (students)	40	36		

¹Effect size equals the difference in treatment means divided by the control standard deviation.

* $p < .05$.

** $p < .01$.

were selected as friends by nonhandicapped students significantly more often than were their counterparts in the comparison schools, $t = 4.33$, $p < .01$. This comparison had an effect size of +.90, nearly a full standard deviation difference (see Table 3).

There were no initial differences between the cooperative elementary school's and the comparison school's learning disabled students' attitudes toward or perceived ability in reading, language arts, or mathematics. Learning disabled students in the cooperative elementary school had higher post-measures of their perceived ability in reading ($t = 2.01$, $p < .05$) and language arts ($t = 2.19$, $p < .05$). The effects sizes for these significant effects are +.26 and +.33, respectively (see Table 5).

Effects for gifted students. Our initial hypotheses did not explicitly predict effects of the cooperative elementary school on gifted students, but, in response to recent discussions about the effects of cooperative learning's effects on gifted students (e.g., Allan, 1991; Feldhusen & Moon, 1992; Slavin, 1991), we also investigated the effects in these data post hoc. Gifted students were identified as the top 10% of students on the standardized achievement pretests. Typically, they were more than a year above grade level. In the comparison schools, the gifted students participated in a twice-weekly pull-out program that focused on enrichment rather than acceleration. The goal

Table 4
Special Education Students: Standardized Achievement Measures, Means, Standard Deviations, and Effect Sizes (Collapsed Across Grades)

Measure	CES		Comparison		F	Effect size ¹
	M	(SD)	M	(SD)		
Pretests						
Reading	-1.31	(.78)	-1.29	(.70)	1.33	-.03
Language	-1.27	(.81)	-1.30	(.65)	3.19	+.05
Mathematics	-.98	(.73)	-.98	(.80)	<1.0	.00
Year 1						
Posttests						
Read. voc.	-1.20	(.81)	-1.38	(.69)	1.28	+.26
Read. comp.	-1.21	(.83)	-1.40	(.65)	1.59	+.29
Lang. mech.	-1.16	(.87)	-1.25	(.79)	<1.0	+.11
Lang. expr.	-1.19	(.85)	-1.45	(.70)	2.59	+.37
Math comp.	-.45	(.70)	-.71	(.78)	2.06	+.33
Math appl.	-.71	(.66)	-.83	(.75)	<1.0	+.16
Year 2						
Posttests						
Read. voc.	-.98	(.86)	-1.48	(.66)	13.83**	+.76
Read. comp.	-1.00	(.96)	-1.53	(.62)	14.39**	+.85
Lang. mech.	-.99	(.96)	-1.21	(.85)	<1.0	+.25
Lang. expr.	-.97	(.97)	-1.55	(.78)	11.41**	+.74
Math comp.	-.10	(.65)	-.54	(.75)	10.77**	+.59
Math appl.	-.28	(.58)	-.53	(.71)	3.98*	+.35
N (students)	40		36			

¹Effect size equals the difference in treatment means divided by the control standard deviation.

* $p < .05$.

** $p < .01$.

of the enrichment program was to extend gifted students' learning to higher levels of understanding through self-directed investigations and problem-solving activities. In the cooperative elementary school, the gifted students did not participate in enrichment activities; instead, they remained in the regular classroom and participated on heterogeneous cooperative learning teams.

The gifted students in the treatment and comparison schools were not significantly different from one another on their initial reading ($F = 1.21$) and language achievement ($F = 2.11$). However, the gifted students in the comparison schools had significantly higher initial math achievement than those in the treatment schools ($F = 4.28$, $p < .05$). Due to the initial differences, an analyses of covariance (ANCOVA) was used to analyze the posttest data, controlling for pretests in reading, language, or mathematics.

The first-year results indicate no significant differences on achievement for gifted students in the cooperative elementary schools as compared to

Table 5
**Special Education Students' Attitudes, Means, Standard
 Deviations, and Effects**

	CES	Comparison	<i>t</i>	Effect size ¹
Attitude toward reading				
Pretest	1.53 (.64)	1.67 (.67)	1.34	-.21
Posttest	1.56 (.68)	1.63 (.70)	1.21	-.10
Attitude toward language arts				
Pretest	2.07 (.70)	2.05 (.80)	<1.0	+.03
Posttest	1.89 (.81)	1.83 (.74)	<1.0	+.08
Attitude toward mathematics				
Pretest	1.30 (.74)	1.27 (.54)	<1.0	+.06
Posttest	1.43 (.75)	1.44 (.59)	<1.0	-.02
Perceived reading ability				
Pretest	1.80 (.68)	1.84 (.77)	<1.0	-.05
Posttest	1.95 (.50)	1.79 (.54)	2.01*	+.26
Perceived language arts ability				
Pretest	1.95 (.70)	2.00 (.77)	<1.0	-.06
Posttest	2.00 (.77)	1.79 (.63)	2.19*	+.33
Perceived math ability				
Pretest	1.73 (.71)	1.67 (.67)	1.48	+.09
Posttest	1.71 (.85)	1.63 (.70)	1.30	+.11
<i>N</i> (students)	40	36		

¹Effect size equals the difference in treatment means divided by the control standard deviation.

* $p < .05$.

** $p < .01$.

those in the comparison schools. After 2 years, the gifted students in the treatment schools had significantly higher achievement in reading vocabulary, $F = 11.06$, $p < .01$; reading comprehension, $F = 12.13$, $p < .01$; language expression, $F = 6.09$, $p < .05$; and math computation, $F = 9.16$, $p < .05$, than did their counterparts in the comparison schools who participated in enrichment pull-out programs (see Table 6). There were no significant differences on measures of language mechanics or math applications. The effect sizes for the significant main effects ranged from +.48 to +.68 standard deviations, larger than the effects for students in general, but similar to those found for special education students.

Initially there were no significant differences between gifted students in the cooperative elementary school and those in the comparison schools on the number of friends they listed. After 2 years in the cooperative elementary school, gifted students reported significantly more friends than did the gifted students in the comparison schools ($t = 2.64$, $p < .01$). The gifted students in the cooperative elementary school averaged 1.5 more friends, an effect size of +.46 (see Table 3).

Table 6
**Gifted Students: Standardized Achievement Measures, Means,
 Standard Deviations, and Effect Sizes (Collapsed Across Grades)**

Measure	CES			Comparison			<i>F</i>	Effect size ¹
	<i>M</i>	(<i>SD</i>)	Adjusted <i>M</i>	<i>M</i>	(<i>SD</i>)	Adjusted <i>M</i>		
Pretests								
Reading	1.47	(.48)		1.59	(.60)		1.21	-.20
Language	1.13	(.53)		1.31	(.68)		2.11	-.26
Mathematics	1.38	(.64)		1.68	(.55)		4.28*	-.54
Year 1								
Posttests								
Read. voc.	1.43	(.39)	1.47	1.30	(.64)	1.25	3.03	+.34
Read. comp.	1.34	(.40)	1.38	1.35	(.57)	1.30	<1.0	+.14
Lang. mech.	1.10	(.51)	1.10	1.14	(.62)	1.13	<1.0	-.05
Lang. expr.	1.12	(.46)	1.13	1.10	(.57)	1.09	<1.0	+.07
Math comp.	1.32	(.63)	1.37	1.24	(.56)	2.69	9.16**	+.32
Math appl.	1.34	(.59)	1.39	1.48	(.55)	1.42	<1.0	-.05
Year 2								
Posttests								
Read. voc.	1.38	(.33)	1.41	1.01	(.66)	.99	11.06**	+.65
Read. comp.	1.14	(.24)	1.15	.86	(.44)	.85	12.13**	+.68
Lang. mech.	1.06	(.48)	1.12	1.10	(.53)	1.06	<1.0	+.11
Lang. expr.	1.16	(.31)	1.22	1.07	(.40)	1.03	6.09*	+.48
Math comp.	1.24	(.61)	1.33	1.05	(.58)	.99	9.16**	+.59
Math appl.	1.30	(.43)	1.37	1.33	(.54)	1.27	<1.0	+.19
<i>N</i> (students)		46			61			

¹Effect size equals the difference in adjusted means divided by the control standard deviation.

* $p < .05$.

** $p < .01$.

The attitude measures indicated no significant pretest differences between gifted students in the cooperative elementary school and those in the comparison schools on their attitudes toward reading, language arts, or mathematics, or their perceived ability in those subjects. After 2 years, gifted students in the cooperative elementary school had significantly higher attitudes toward language arts ($t = 2.09$, $p < .05$) and higher perceived ability in language arts ($t = 3.55$, $p < .01$). Their attitudes toward school subjects and perceived ability were rated nearly a half to two-thirds of a standard deviation higher on these two measures (effects sizes = +.48 and +.68, respectively; see Table 7).

Discussion

Because the experimental program had so many components, it is difficult to ascribe the program's outcomes to any single element. However, the

Table 7
**Gifted Students' Attitudes, Means, Standard Deviations,
 and Effects**

	CES		Comparison	<i>t</i>	Effect size ¹
Attitude toward reading					
Pretest	1.60	(.75)	1.69 (.62)	1.33	−.14
Posttest	1.95	(.69)	1.90 (.65)	<1.0	+ .08
Attitude toward language arts					
Pretest	1.78	(.70)	1.81 (.75)	<1.0	−.04
Posttest	2.05	(.83)	1.67 (.79)	2.09*	+ .48
Attitude toward mathematics					
Pretest	1.45	(.61)	1.49 (.64)	<1.0	−.06
Posttest	1.52	(.54)	1.50 (.58)	<1.0	+ .03
Perceived reading ability					
Pretest	1.51	(.59)	1.55 (.61)	<1.0	−.07
Posttest	1.52	(.61)	1.50 (.51)	<1.0	+ .04
Perceived language arts ability					
Pretest	1.75	(.55)	1.79 (.54)	<1.0	−.07
Posttest	1.85	(.75)	1.33 (.77)	3.55**	+ .68
Perceived math ability					
Pretest	1.30	(.67)	1.28 (.54)	<1.0	+ .04
Posttest	1.26	(.58)	1.27 (.51)	<1.0	−.02
<i>N</i> (students)	46		61		

¹Effect size equals the difference in treatment means divided by the control standard deviation.

* $p < .05$.

** $p < .01$.

results of this study support the hypothesis that cooperative learning can be the primary mode of instruction and—when integrated with effective instruction in reading, language arts, and mathematics and with changes in school organization, peer coaching, mainstreaming, and other elements—can be effective in producing higher student achievement. After the first year, the effects were relatively small, and the only significant differences were on measures of reading vocabulary achievement, which indicated that students in the cooperative elementary school outperformed those in the traditional schools. The first year results in reading, to some degree, replicate earlier findings of the positive effects for the CIRC program (see Stevens et al., 1987). The lack of significant positive effects in math and language arts is likely to be due to the later implementation of these components, which were gradually phased in over the course of the year. As a result, the students received the treatment in language and math for much less than a year, and it is possible that students have to experience the programs longer for them to have a strong impact on standardized achievement test scores.

After the second year, the students in the cooperative elementary schools clearly outperformed their peers in the more traditional schools on standard-

ized measures of reading vocabulary, reading comprehension, language expression, and mathematics computation. The effect sizes, which ranged from $+ .21$ to $+ .29$ standard deviations, suggest that these differences were educationally significant—particularly, considering that the differences were obtained on standardized achievement tests, which typically are difficult to influence due to their limited overlap with the curriculum and their strong correlation with general ability. The pattern and the magnitude of these effects replicate the findings in earlier studies for the CIRC and TAI programs (see Slavin et al., 1984; Stevens et al., 1987; Stevens & Slavin, 1995).

These results are important because they are the result of a long-term implementation of cooperative learning. The positive effects suggest that multifaceted cooperative learning programs like TAI and CIRC can maintain their effectiveness over time and that the effectiveness is not likely to be due to the novelty of cooperative learning or to the Hawthorne effect, as has been suggested (Bossert, 1989). This shows that multifaceted instructional programs that use state-of-the-art instruction and curriculum with cooperative learning classroom processes can produce and maintain significantly higher achievement.

The students in the cooperative elementary school listed significantly more friends than did students in the more traditional schools, indicating that there were better peer relations in the cooperative elementary schools, a result that replicates that of previous research in cooperative learning (cf. Johnson & Johnson, 1989; Sharan, 1980; Slavin, 1994). Unlike typical classroom instruction, during which students tend to have only casual and rather superficial contact, cooperative learning processes have students working together to achieve a common goal. This produces more meaningful interactions between the students and a sense of positive interdependence. As these work groups change over time, the students collaborate with a variety of students in the class, leading to better peer relations in the class and an increase in students' friendships.

Typically, cooperative learning results in more positive student attitudes in the subjects where cooperative learning processes are used (cf. Slavin, 1994). Unlike the previous research, this study did not find that students had enhanced attitudes toward subjects where cooperative learning was used. The lack of significant findings on students' attitudes and on most of the measures of their perceived ability may be due to errors in measurement. The attitude and perceived ability measures had few (3) items, and alpha reliabilities were less than .40. However, students in the cooperative elementary school did have more positive perceptions of their ability in reading and language arts than did the students in the more traditional schools. It is difficult to determine which component of the programs produced this effect on students' perceptions of their ability. The result may in part be due to the cooperative learning processes that reduce the competition in the classroom, for competition has been found to lower most students' perceptions of their ability (Deutsch, 1949). It may be due to the support structure that cooperative learning provides, which may keep students from being

frustrated and hence feeling less successful when they engage in their learning activities. The result may also be due to the curricula used in reading and language arts. Students may perceive their ability as being higher because the instructional activities build competence the students can easily see. The best example of this would be in the case of the writing, where students spent much of their time writing, revising their writing, and critiquing their peers' writing. It is possible that the students more readily perceived the development of their writing ability through the frequent writing activities and that this perception was reflected in their ratings of their perceived ability. Finally, the positive results may be due to some combination of these and other elements of the model.

Mainstreamed learning disabled students. The first-year achievement data show small and nonsignificant advantages for learning disabled students in the cooperative elementary schools. This may in part be due to the difficulties encountered in scheduling fully mainstreamed services for many of these students during the first year. Before the beginning of the second school year, we collaborated with the principals and special education teachers to schedule special education services for learning disabled students when the entire school schedule was being developed. This alleviated what had previously been a roadblock to mainstreaming the learning disabled students and special education teachers into regular classes for instruction in reading and mathematics.

The results after the second year of the study support the hypothesis that the well-structured, cooperative learning programs used in the cooperative elementary school can be vehicles to enhance a school's ability to successfully mainstream learning disabled students into the regular classroom. The achievement effect sizes for learning disabled students, which ranged from a third to more than three quarters of a standard deviation, suggest that learning disabled students have much to gain by being mainstreamed in these kinds of programs. However, these results are dependent on (a) the learning disabled students' being integrated into heterogeneous learning teams in classrooms, (b) the cooperative learning programs' using group goals based on individual accountability, and (c) the special education teachers' being scheduled to provide additional instruction and support to the learning disabled students in the regular classroom.

One advantage of cooperative learning is that it provides a structure that allows students to help manage the classroom, which evolves from the positive interdependence created within the learning teams. When the students take more responsibility, it frees teachers from time that must be spent on managerial tasks so they can provide more individualized instruction to students who need it. This gives teachers more instructional flexibility to accommodate the increased classwide heterogeneity inherent in mainstreaming (Johnson & Johnson, 1989; Slavin, 1994). Additionally, students become an instructional and motivational resource in the classroom so that students who need help can rely on support and feedback from their peers,

providing another mechanism for accommodating students with diverse abilities.

The instructional support for learning disabled students in the cooperative elementary school is further enhanced by the use of an in-class model of support services, where the special education teacher goes into the regular education classroom and team teaches with the classroom teacher. In this way, the learning disabled student continues to receive the additional instruction from the special education teacher while at the same time getting the social and motivational benefits of being in the regular classroom.

The students in the cooperative elementary schools also displayed much better social acceptance of mainstreamed learning disabled students. A learning disabled student in the cooperative elementary school was likely to be chosen as a friend by a nondisabled student 50% more often than were his or her peers in the more traditional schools. This is a particularly important result because one of the goals of mainstreaming is to promote more social acceptance of students who have special needs. However, research has suggested that the opposite effect may often be true. Mainstreamed handicapped students typically are not well integrated into the class socially or academically and as a result are poorly accepted by their peers and often ostracized (Bryan, Bay, Lopez-Reyna & Donahue, 1991; Gottlieb, Corman, & Curci, 1984; Reid, 1984).

Perhaps the critical element in the social acceptance of mainstreamed students is that cooperative learning creates an environment of positive interdependence within the teams, where students depend on one another and where all must succeed in order for the group and any one member to succeed. This makes contact between learning disabled and nondisabled students on the cooperative learning teams meaningful and of more equal status. Allport (1954) found that prejudice actually increased when different racial groups had more frequent superficial contact, much like what may be the case in some mainstreamed classrooms. On the other hand, prejudice decreased when individuals had equal status contact in situations where they could develop a feeling of positive interdependence (Allport, 1954) like that found in well-structured, cooperative learning. The results of this study suggest that true mainstreaming goes beyond academics and that disabled and nondisabled students also need classroom environments that will promote social development and acceptance in an environment of meaningful contact focused on attaining a common goal.

Cooperative learning and gifted students. While we had no a priori hypotheses about the effects of the cooperative elementary school model on gifted students, there was the implicit expectation that the elements of the model would be beneficial for all students, including the gifted. In particular, recent discussions in journals suggest that the effect of cooperative learning on gifted students is an important question. Critics of cooperative learning have suggested that when gifted students are in cooperative learning and take time out of their academic tasks to explain content to their classmates it reduces the achievement, motivation, and attitudes of gifted students (Allan,

1991; Feldhusen & Moon, 1992). Cooperative learning is claimed to result in a slower instructional pace for gifted students, which in turn contributes to lower achievement than that they might have obtained otherwise (Robinson, 1990). Similarly, Linnemeyer (1992) suggests that in heterogeneous cooperative groups the gifted students have no one who can adequately give them the critical feedback assumed to produce higher achievement. On the other hand, researchers have suggested that cooperative learning in heterogeneous groups can be beneficial for gifted students, particularly if teachers are well-trained and the process is implemented well (Gallagher, 1991; Mills & Durden, 1992; Slavin, 1991).

The results from this study suggest that well-structured cooperative learning is not detrimental to the achievement of gifted students and, in fact, can produce significant and substantial positive effects on gifted students' achievement. Gifted students in this study who worked in heterogeneous cooperative groups in a variety of content areas had much higher achievement than similar students in the comparison schools who received enrichment programs twice a week and did not participate in cooperative learning. These results corroborate previous evidence about the benefits of cooperative learning for gifted students (Johnson & Johnson, 1989; Slavin, 1994; Webb, 1985). Gifted students seem to benefit from being the one who provides elaborate explanations to classmates, a process that is important to learning both at a theoretical level (Brown & Palincsar, 1989; Collins, Brown, & Newman, 1989; Vygotsky, 1978) and in practice (Webb, 1985).

Limitations and Future Research

This study did not evaluate all of the components of the cooperative learning program. Instead, it focused on using cooperative learning in a variety of content areas and mainstreaming learning disabled students. Components, such as peer coaching and cooperative planning among teachers and between the teachers and principal, were not specifically addressed. The authors felt that, for the purposes of this study, these factors primarily served a facilitative role in promoting the changes in curricula, instructional processes, and school and classroom organization. It is impossible to determine the impact of any of these components from the data presented. Similarly, each of the cooperative learning programs (e.g., TAI, CIRC) changed group processes, curricula, and reward structures, and the relative effects of parts cannot be disentangled in this study. Rather, such effects need to be the goal of future component analysis studies (cf. Stevens, Slavin, & Farnish, 1991).

The study's assignment procedures raise the possibility of selection bias. The experimental schools were selected only if their teachers voted to implement the program. Selection bias would have been a more serious problem if we had compared these schools to schools that considered and rejected the program. In fact, there were only two schools we approached that did not choose to participate, so it is reasonable to assume that, given the chance, the control schools would quite likely have chosen to participate.

The only way to have removed selection bias would have been to randomly assign schools to treatments, something that, as a practical matter, was simply infeasible. Our use of HLM gives us a statistical penalty for having students grouped within schools whose characteristics may influence outcomes. Also, if better schools had been selected for the experimental group, it would presumably have been reflected in the premeasures, which, in fact, favored the control schools.

Another potential limitation of this study is that we did not quantitatively measure the implementation of the various cooperative learning programs or other elements of the cooperative elementary school. Instead, our goal was to coach teachers to improve the quality of their implementation. Clearly, there was a significant amount of natural variation across teachers in the cooperative elementary school, as there is in all schools and all programs, which to some degree limited the generalizability of these results. Similarly, the teachers in the cooperative elementary school were volunteers, which suggests that these results cannot be generalized to situations where the model is mandated.

The fact that standardized achievement tests were our only measure of student learning is somewhat limiting. The stability, multiple-choice format, and lack of overlap with the curricula taught inherent in these tests were disadvantages. However, we felt that the disadvantages were more than balanced by the advantages of higher reliability and the fact that the tests were generic. They did not measure specific curriculum content; hence, no specific curricular advantage was given to either the cooperative elementary school or the comparison schools.

Finally, the comparison of gifted students in the cooperative elementary schools to students in gifted enrichment programs may not be generalizable to all forms of gifted education. Clearly, enrichment programs differ from accelerated programs for gifted students, as well as from one another, making generalization to other programs limited. However, the results do show that well-structured, cooperative learning can be beneficial for gifted students.

Any study of a multifaceted program leaves many questions unanswered, and this one is no exception. Perhaps the most evident question revolves around the relative impact of each of the elements of the cooperative elementary school in producing the various effects on achievement, social relations, and students' attitudes. Only through component analyses of this model can one begin to disentangle the elements and their effects. Similarly, there remain questions about the nature of the student dialogue and classroom processes in the cooperative learning setting. It may be particularly important to understand these processes in terms of how learning disabled or other below-average students participate in collaborative dialogue and how that dialogue can help or hinder their development.

The duration of the study greatly reduces the potential for the Hawthorne effect to have produced the observed differences in students' achievement. Hawthorne effects are primarily novelty and attention effects. In a brief study, these effects could be consequential. However, it does not seem likely that

novelty and attention could sustain outstanding efforts in all major academic subjects for 2 years, as in this study. If the outcomes were due to Hawthorne effects, the entire effect would be expected to be seen during the first months. Yet in this study more gains were seen in the second year than in the first (see Table 1). This is more consistent with the argument that teachers were getting better at a new method than that students were profiting from a short-term burst of enthusiasm on the part of their teachers. For the students themselves, cooperative learning would have been seen as standard operating procedure long before the end of the 2 years.

While further research is needed to understand the effects of this program, the results of this study do suggest that cooperative learning can serve as a basis for school restructuring and can produce important benefits for a wide range of students. This study is the first and only evaluation of a cooperative elementary school. It is not merely another study of cooperative learning; it is the only study to evaluate cooperative learning as the focus of schoolwide change, the only study to evaluate cooperative learning in many subjects at once, and one of the few to show the effects of cooperative learning over a multiyear period.

Notes

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^aEffect size is the mean difference between the experimental and the control groups divided by the control group's standard deviation.

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